

Sasha Lopoukhine

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EDUCATION

University of Cambridge
PhD in Computer Science supervised by [Tobias Grosser](#)

Cambridge, UK
Jan 2023 — Present

Developing assembly-level MLIR representations and compiler tooling for linear-algebra micro-kernels on AI accelerators.

CGO 2025: [A Multi-level Compiler Backend for Accelerated Micro-kernels Targeting RISC-V ISA Extensions](#) Developed an MLIR-based linear algebra micro-kernel backend that targets ETH’s Snitch Core, yielding 95% FPU utilization and 94% of theoretical maximum throughput for our chosen kernels. Neural networks compiled via this micro-kernel outperform output of LLVM by >20x.

CGO 2025: [Sidekick compilation with xDSL](#) Leading development of xDSL, a Python-native, MLIR-compatible compiler framework used to prototype and ship compiler infrastructure in Python. Now at 1M+ PyPI downloads and 125+ contributors; production use includes quantum compilers (Riverlane, Xanadu, Eclipse) and Google LiteRT; research integrations include Devito and PSyclone. Featured in [GitHub Maintainer Month](#) added to the CPython performance benchmark suite; cited in PEP 747.

Supervised four BA and one MA dissertation projects, one of which won silver at the [CGO 2024 student research competition](#).

University of Cambridge
BA in Computer Science - II.i

Cambridge, UK
2012 — 2015

Received Bob Diamond prize for Computer Science.

EMPLOYMENT HISTORY

Student Investment Partner (promoted from Venture Fellow)
Creator Fund

Oct 2024 — Present
Cambridge, UK

Sourced Cambridge-based startups and conducted technical due diligence for investment decisions.

Co-Founder & Chief Engineering Officer
Cub3

Sep 2021 — Jan 2022
Los Angeles, CA

Co-founded Cub3; led engineering from zero to MVP of a Web3 fan-engagement platform.

Software Engineer
Apple

Mar 2018 — Sep 2021
Cupertino, CA

[Curated Library](#) – iOS 13 feature shipped to **1B+ devices**.

- Built the [Years/Months/Days](#) timeline from photo metadata. **Patent:** US20200356227A1 – **Curated Media Library**
- Profiled and optimized Years/Months/Days timeline computation to ensure low-latency processing when taking photos.

[On-disk graph](#) – Implemented a graph database for Memories data (~1000x prior capacity).

- Unlocked on-demand availability of all Memories, automatic song selection, and next memory suggestion.
- Led the migration of existing features with minimal regressions, reducing benchmark runtime **9×** (90 → 10 min).

[Next memory suggestion](#) – Developed a recommendation engine for Memories.

[Apple Music in Memories](#) – Led feature extraction from the Memories DB for music recommendation.

- Mentored the team on Swift and software architecture.

[Keyboard extension](#) – system-wide iOS keyboard suggesting information from the Snips Personal Knowledge Graph.

- Integrated email, calendar, and contacts for real-time context as the user typed.
- Developed a binary-encoded language model for auto-completion and next-word prediction; loaded in milliseconds.

Snips Platform – Labs - prototyped natural language parsers and voice assistants on iOS.

- Built iOS voice assistant handling contextual queries (e.g. “Order cab to restaurant I went to last week.”) via CKY parsing.
- Built query gap-filling engine for incomplete voice queries (e.g. “Which restaurant? **A** or **B**?”).
- Prototyped voice UIs for embedded devices (speakers, appliances).

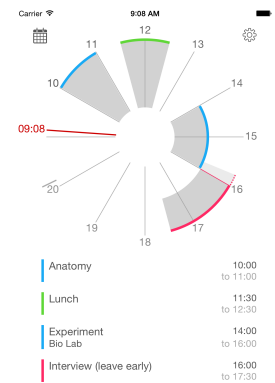
Co-Founder & CEO

Blue Tatami

Co-founded Blue Tatami, an iOS personal productivity startup.

- Designed and shipped Dodeka, a 12-hour watch-face calendar app, from scratch.
- Grew to **6,000+** users with no paid advertising.

2014 — 2016
Cambridge, UK



PROJECTS

2017: [Simple Datagram Protocol](#) | [Article](#) – Bypassed MFi restrictions by using BTLE to configure Raspberry Pis from iOS.

2015: [BioSwift](#) – First open-source framework in Swift to handle gene data.

PRESENTATIONS

[2023 EuroLLVM - Prototyping MLIR in Python](#)

[2023 LLVM Dev Mtg - Compiler backend design with MLIR](#)

[2024 EuroLLVM - Teaching MLIR concepts to undergraduate students](#)

[2024 LLVM Dev Mtg - Quidditch: An End-to-End Deep Learning Compiler for Occamy using IREE & xDSL](#)

TEACHING

Taught introductory MLIR sessions at the [2025 MLIR Winter School](#), [2025 MLIR Summer School](#).

SKILLS

- **Programming Languages:** Python, Swift, Objective-C, C/C++, JavaScript/TypeScript, Assembly
- **Languages:** Fluent in English, Russian, French